

GOWTHAM KANDASAMY

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 7 years of experience in backend architecture, workflow automation, and technical leadership. Proven record reclaiming business capacity and delivering **\$13k/year** infrastructure cost savings. Expert across Python and Java ecosystems with hands-on delivery of **RAG platforms**, vector search, FastAPI microservices, and React dashboards. Leverages Cursor AI, Model Context Protocol (MCP), and containerized deployments to accelerate feature velocity.

CORE COMPETENCIES

AI & Architecture	Retrieval-Augmented Generation (RAG), Vector Embeddings, pgvector, Google Gemini, Model Context Protocol (MCP), Cursor AI, Ollama, Docling
Languages & Frameworks	Python (FastAPI, SQLAlchemy, SQLAlchem, Hamilton), Java 17 (Spring Boot, Hibernate), TypeScript, React 19, Vite, Tailwind CSS, Streamlit, PostgreSQL, MySQL, ChromaDB, Alembic
Automation & DevOps	Docker, Docker Compose, Kubernetes (EKS), Helm, AWS (S3, Secrets Manager), CI/CD, RPA (UiPath), Selenium, PyMuPDF, PDFlib TET, WebSockets, Google Apps Script
Security & Integrations	JWT, bcrypt, Okta, Azure AD, Dropbox API, Microsoft Teams Webhooks, OpenCSV, REST APIs

PROFESSIONAL EXPERIENCE

Senior Software Engineer Perch Energy (Helios Arcadia New Energy)	Remote	Mar 2025 – May 2026
- Architected a centralized automation hub that cut manual processing time by 85%, streamlining the critical ingestion layer for 18 global utilities across 198 concurrent projects. - Implemented a Hamilton-based DAG methodology for dynamic data processing, managing scalable subscriber profiles for over 60,000 accounts. - Accelerated engineering velocity by 50% by designing context-aware Cursor rules and local tool mappings that automated codebase indexing for high-precision utility components.		
Senior Software Engineer Arcadia (Urjanet, an Arcadia enterprise)	Chennai, India	May 2022 – Mar 2025
- Engineered a high-performance Java 17 contract platform (CAE) with a three-stage Auto-QC engine for structural PDF text-layer diffing, reclaiming 3 FTE capacities daily. - Won the 2024 Corporate Hackathon by migrating legacy toolsets to open-source PyMuPDF, yielding \$13k/year in recurring operations cost savings. - Built the flagship TDS engine handling 140+ global LSEs using Selenium and PDFlib TET for automated text extraction at scale. - Selected to pioneer technical screening; interviewed, hired, and onboarded high-performance engineering squads to scale production.		
Software Engineer Urjanet Energy Solutions	Chennai, India	May 2019 – May 2022
- Standardized regional layout schemas for 50+ utilities using Java and Hibernate backend parsing components. - Automated JIRA triage and metrics analysis with REST API integrations inside the internal anomaly detector platform (PAD). - Published custom Google Apps Script functions to automate sheet tracking across cross-functional operational squads.		

KEY TECHNICAL PROJECTS

ClaimsAuditor AI — Enterprise Health Insurance Claims Auditing Platform

FastAPI · React 19 · TypeScript · PostgreSQL · pgvector · Gemini 2.5 Flash · Docling · Docker Compose · JWT · WebSockets · Alembic | Personal Project 2026

- Architected a full-stack AI auditing platform: policy PDF ingestion via Docling, heading-aware chunking, Gemini embeddings (3072-dim) stored in pgvector, and a React 19 dashboard with live WebSocket progress (Policy Ingestion, Claims Desk, Audit Console, RAG Chat views).
- Built a Gemini 2.5 Flash cognitive audit engine that compares claim vs. policy Markdown and returns structured APPROVED / FLAGGED / DENIED verdicts with policy citations and violation severity.
- Delivered JWT multi-tenant auth, user-scoped repositories, Alembic migrations, async 202-Accepted background pipelines, and real-time WebSocket status streams.

AI Personal Habit Tracker & Assistant Coach

FastAPI · PostgreSQL · Streamlit · FastMCP (SSE) · ChromaDB · Ollama | Personal Portfolio 2026

- Containerized habit tracker with an MCP Server over SSE for deterministic agent-driven database operations and a local RAG layer (ChromaDB, all-minilm:v2, 384-dim) injecting historical trends into the system prompt.
- Repository hooks keep relational data and vector memory in sync; persistent chatbot sidebar available across all Streamlit views.

EDUCATION & HONORS

B.E. Electronics & Communication Engineering — Karpagam College of Engineering (Anna University) | 2015–2019 | Score: 86.63%

Honors: Best Project of the Year (CAE) | "It's My Ship" Award | Multiple quarterly engineering excellence milestones for architectural innovations.